You will apply the knowledge on parameter management acquired in the previous session to the physical activity planner problem in the coursework brief. This will give you a practical understanding of the tuning and control techniques we discussed. You will document the ACTIVEA parameter management experiments and use the findings to complete your parameter management section of your report and presentation of your coursework assessment.

TASK 1: READING — 10 MIN

Cover chapter 9 of the textbook used in the previous session:

A. E. Eiben and James E. Smith. 2015. Introduction to Evolutionary Computing (2nd. ed.). Springer Publishing Company, Incorporated - full text available at https://read.kortext.com/reader/pdf/406661/C1

As you read, make notes on the performance measures commonly used to evaluate the quality of evolutionary models (section 9.2) and on the various ways evolutionary algorithms may be assessed in practice (section 9.4). In your group, discuss and collate your notes to reach a common understanding of (1) the performance measures you will be applying to ACTIVEA and (2) the way in which you will be calculating those measures to reflect best practice.

TASK 2: REVISION — 10 MIN

Revisit the code you wrote in the week 1 practical. Identify the key parameters for the parameter management experiments (consult the summary you developed in the previous session).

TASK 3: EXPERIMENTS — 20 MIN

Tune and manage your algorithm parameters by applying the techniques covered in the previous session (consult your group's summary). For each parameter configuration, run the code 10 times and record the results in a table. The table should have two columns: one that lists the parameter configurations and one that shows the corresponding calculations for all algorithm performance measures in section 9.2 of the textbook. Make sure to follow the good practice in section 9.4 of the textbook.

Note. *10 repetitions of the experiment are not sufficient for statistical significance. This task is set to be doable in the classroom. You will need to complete the full set of at least 30 experiment repetitions for the coursework in order to be able to make claims that are supported by statistically significant evidence.*

TASK 4: REFLECTION — 10 MIN

Revisit and interpret your experimental results. Add the results and interpretations to the presentation started in the previous session.